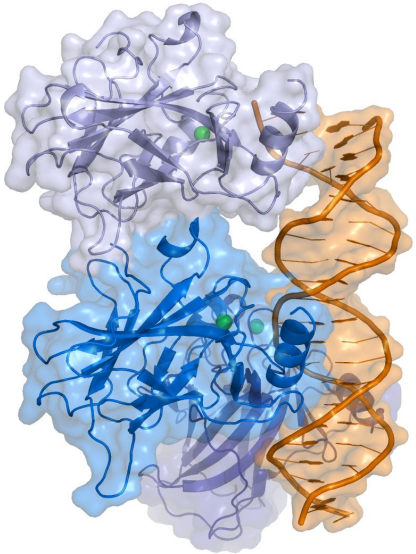
Design of p53-Targeting Peptides Using Language Model-Based Sequence Generation

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Overview

Problem Statement

- Frequent TP53 mutations impair p53's role in cancer prevention as the main "guardian of the genome.
- p53's disordered structure lacks binding pockets for small molecules.
- MDM2-mediated ubiquitination (E3 ubiquitin ligase) tightly regulates p53, complicating stabilization.
- Novel strategies necessary to selectively stabilize and activate p53.



Overview

Problem Statement → *Methods*

- Use computational methods to generate candidate peptides
- Screen top candidates using perplexity score
- In vitro screening with phage/yeast display*
- Express candidate binder in bacteria and purify
- Use ELISA assay with p53 and compare binding affinity

Name	Sequence
p53_pMLM_001	SDEPLLGGMPRT
p53_pMLM_002	SDTPLYSGCPRT
p53_pMLM_003	GDYYPLSGCLGT
p53_pMLM_004	DDEPRYSGMPRY
p53_pMLM_005	SDYPLYGPMCRY
p53_pMLM_006	SDTPLYGLPLRT
p53_pMLM_007	SDYEPYGPPCRR
p53_pMLM_008	GPYYPLSGMLRR
p53_pMLM_009	GDYYPLPGPLRR
p53_pMLM_010	DDYPLYGGPLGR

*Not performed due to time constraints

Overview

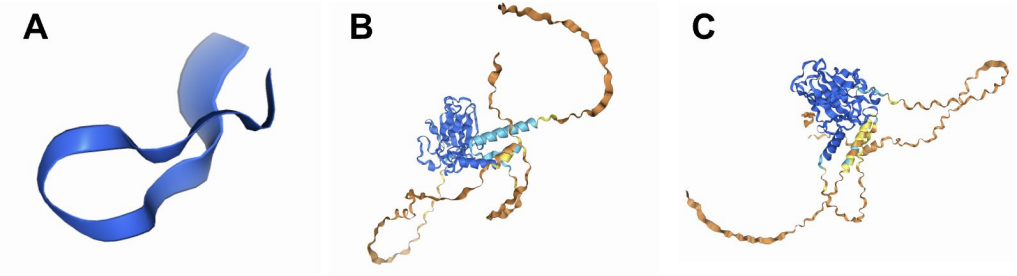
Problem Statement → *Approach* → *Results*

- Binding affinity of binder to control and PBS in nM range
- No statistical significance
- High data variability

Computational Design

- pepMLM used to generate binder sequences
- Top sequence selected for *in vitro*
- AlphaFold used to co-fold p53 with binder
- ipTM score of 0.22
  - Intrinsic disorder in binder

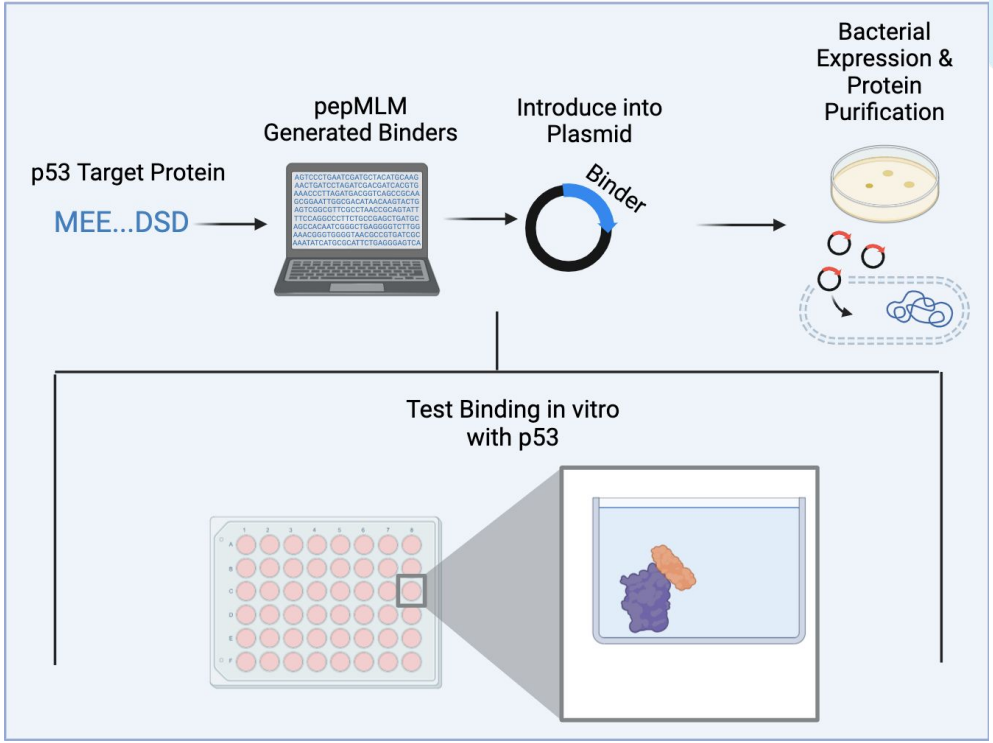
Binder Sequence	Pseudo-Perplexity
SDEPLLGGMPRT	7.70
SDTPLYSGCPRT	8.54
GDYYPLSGCLGT	8.90
…	…



A) binder B) p53 C) binder:p53

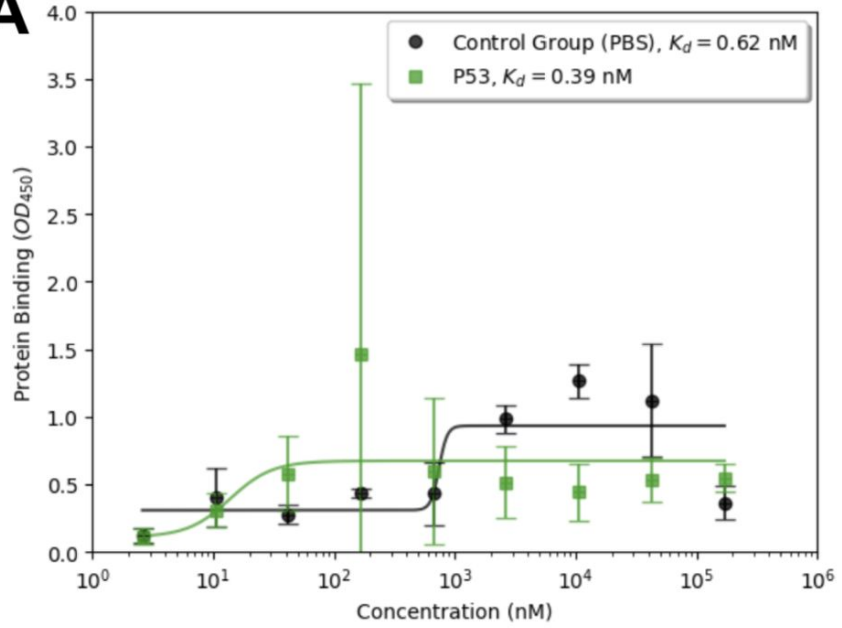
Experimental Methods

- Plasmid generation
  - Gibson Assembly of insert
  - *E. coli* transformation
- Protein expression and purification
- Binder biotinylation
- ELISA binding assay between candidate peptide and p53 protein

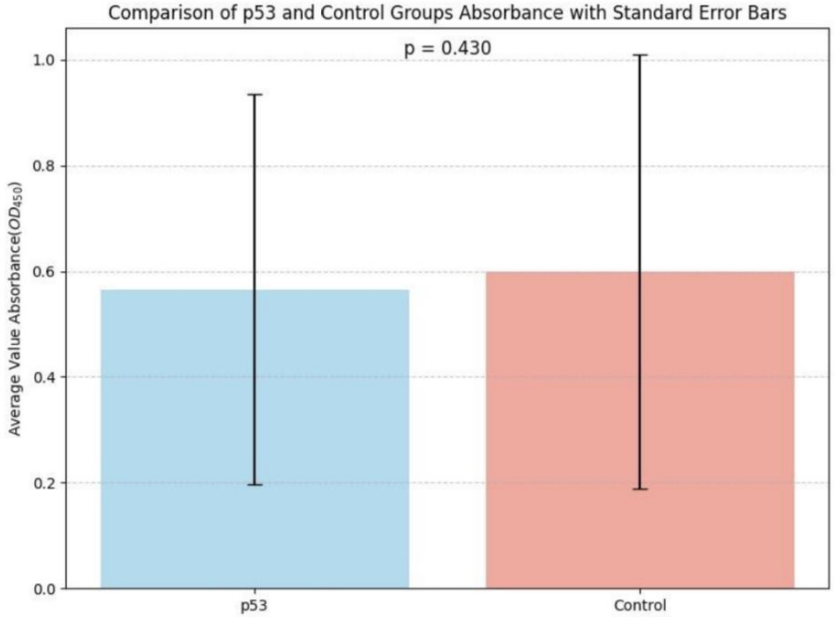


Results

A


Control Group (PBS), $K_d = 0.62$ nM
P53, $K_d = 0.39$ nM
Protein Binding (OD_{450})
Concentration (nM)
4.0
3.5
3.0
2.5
2.0
1.5
1.0
0.5
0.0
10^0
10^1
10^2
10^3
10^4
10^5
10^6

B


Comparison of p53 and Control Groups Absorbance with Standard Error Bars
p = 0.430
Average Value Absorbance(OD_{450})
1.0
0.8
0.6
0.4
0.2
0.0
p53
Control

Conclusions

- Sequence-based deep learning models have the potential to overcome the limitations of structure-based approaches
- Pipeline of *in silico* design followed by *in vitro* & *in vivo* testing can be used to evaluate novel candidate therapeutic peptides
- Nonspecific binding is a major hurdle to successful targeting using these *de novo* binders

Future Directions

- **Refine and Expand Computational Approaches:** Integrate additional sequence-based and structure-informed models (e.g., RFDiffusion) to improve peptide design and reduce nonspecific binding.
- **High-Throughput In Vitro Screening:** Employ phage or yeast display libraries to validate computationally predicted sequences, identify top binders, and optimize binding affinity and specificity.
- **Biophysical Characterization and Structural Insights:** Use advanced techniques (e.g., NMR, cryo-EM) to elucidate peptide-p53 interaction mechanisms and guide rational sequence modifications.
- **Functional and Cellular Assays:** Evaluate lead peptides in cellular models to assess their ability to stabilize p53, enhance downstream signaling, and ultimately inhibit cancer cell proliferation.

Thank you!

Q&A